



GUJARAT TECHNOLOGICAL UNIVERSITY

Shree Swami Atmanand Saraswati Institute of Technology

A

Project Report on

SMART HOUSE PRICE PREDICTION SYSTEM

Under subject of

Python for Data Science (BE05000231)

B.E. 3rd Year, Semester – V

COMPUTER ENGINEERING

Under the Guidance of:

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CERTIFICATE

This is to certify that the Project entitled “Smart House Price Prediction System” has been prepared by **Khush Gopani (240760107057), Ved Goyani (240760107063), Manan Hirpara (240760107069), Daksh Bhalala (250763107003), Manav Bhoi (250763107004), Gadhiya Vaidik (250763107008)** in the subject of **Python for Data Science (BE05000231)** under my guidance in partial fulfillment of the degree of **Bachelor of Engineering in Computer Engineering** Department (5th Semester) of Gujarat Technological University, Ahmedabad during the academic year 2026–2027.

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Abstract

House price prediction is a useful application of machine learning because property prices depend on many characteristics such as living area, overall quality, location, year of construction, garage facilities, and other property attributes. Estimating prices manually can be difficult when a large number of properties must be evaluated.

In this microproject, a simple machine learning system is developed to predict residential house prices using the Ames Housing dataset. The project follows a straightforward data science workflow consisting of data loading, basic exploratory data analysis, missing-value handling, categorical encoding, model training, evaluation, and final prediction.

The latest implementation intentionally uses only one machine learning algorithm, the Random Forest Regressor. No custom feature engineering, ensemble learning, stacking, or cross-validation is used. This makes the system easier to understand, faster to execute, and suitable for demonstrating the basic machine learning workflow.

The model is evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). The notebook also displays sample actual and predicted house prices and generates a submission.csv file containing predictions for the test dataset.

Keywords: House Price Prediction, Machine Learning, Random Forest, Regression, Exploratory Data Analysis, Ames Housing Dataset.

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1. Problem Statement

Determining an appropriate selling price for a house is difficult because property prices are influenced by many factors simultaneously. Two houses with similar sizes can have different prices because of differences in quality, location, age, garage facilities, and other characteristics.

The problem addressed in this microproject is to develop a simple machine learning model that learns the relationship between available house characteristics and their sale prices.

The system should:

1. Load the training and testing data from the Ames Housing dataset.
2. Explore the dataset using basic exploratory data analysis.
3. Handle missing values without manually creating domain-specific features.
4. Convert categorical information into numerical form so that it can be used by a machine learning algorithm.
5. Train a single Random Forest regression model.
6. Evaluate the model using RMSE and R^2 on a validation set.
7. Generate house price predictions for the test dataset.
8. Save the final predictions as a CSV file.

The focus is deliberately kept on a simple and understandable implementation rather than a complex multi-model ensemble.

2. Objectives

The main objective of this microproject is to implement a simple and reproducible machine learning workflow for house price prediction.

The specific objectives are:

1. **Data Loading:** Load the Ames Housing training and testing datasets.
2. **Exploratory Data Analysis:** Examine the first few records, missing values, descriptive statistics, and the distribution of house prices.
3. **Basic Data Preprocessing:** Fill missing numerical values using the median and missing categorical values using the mode.
4. **Categorical Encoding:** Convert categorical variables into numerical variables using one-hot encoding.
5. **Model Training:** Train one Random Forest Regressor with 50 trees.
6. **Model Evaluation:** Measure performance using RMSE and R^2 .
7. **Sample Prediction:** Display actual and predicted prices for a small validation sample.
8. **Final Prediction:** Predict prices for the test dataset and save them in submission.csv.

3. Literature Survey

Machine learning based house price prediction has been studied using statistical regression and tree-based algorithms. The Ames Housing dataset is commonly used for demonstrating regression techniques because it contains a variety of numerical and categorical property characteristics.

3.1 Hedonic Pricing Approach

The hedonic pricing approach considers a property as a collection of characteristics. Features such as size, location, quality, number of rooms, and age can contribute to the overall value of a house. Traditional regression methods can be used to estimate the relationship between these characteristics and price.

3.2 Ames Housing Dataset

The Ames Housing dataset was introduced as an alternative dataset for regression projects. It contains residential property sales from Ames, Iowa and provides numerous explanatory attributes describing the properties.

The dataset is useful for machine learning experiments because it contains both numerical and categorical variables and also includes missing values that require preprocessing.

3.3 Random Forest Regression

Random Forest is an ensemble tree-based regression algorithm that builds multiple decision trees and combines their predictions. Each tree is trained using randomized samples and feature selections. Combining the trees can provide more stable predictions than relying on a single decision tree.

In this project, Random Forest is selected as the only predictive model because it can capture non-linear relationships and can work effectively with a mixture of numerical and encoded categorical features.

3.4 Summary

The literature provides several possible approaches for house price prediction. However, the present microproject intentionally uses a simple Random Forest implementation so that the complete workflow remains easy to understand and execute.

4. Dataset Description

The project uses the Ames Housing dataset. The dataset contains residential property records from Ames, Iowa.

4.1 Dataset Overview

The commonly used Kaggle version contains:

- **Training dataset:** 1,460 rows.
- **Testing dataset:** 1,459 rows.
- **Training data:** Contains the target variable SalePrice.
- The dataset contains numerical and categorical property attributes.

4.2 Types of Features

The dataset contains information related to:

- Living and basement areas.
- Lot size and property dimensions.
- Number of rooms and bathrooms.
- Overall material and finish quality.
- Construction and remodeling years.
- Garage and basement characteristics.
- Neighborhood and building type.
- Other property-related categorical attributes.

4.3 Target Variable

The target variable is SalePrice, representing the sale price of a residential property. The model learns a mapping from the available house characteristics to this target value.

4.4 Preprocessing

Missing numerical values are filled using the median value of the corresponding column.

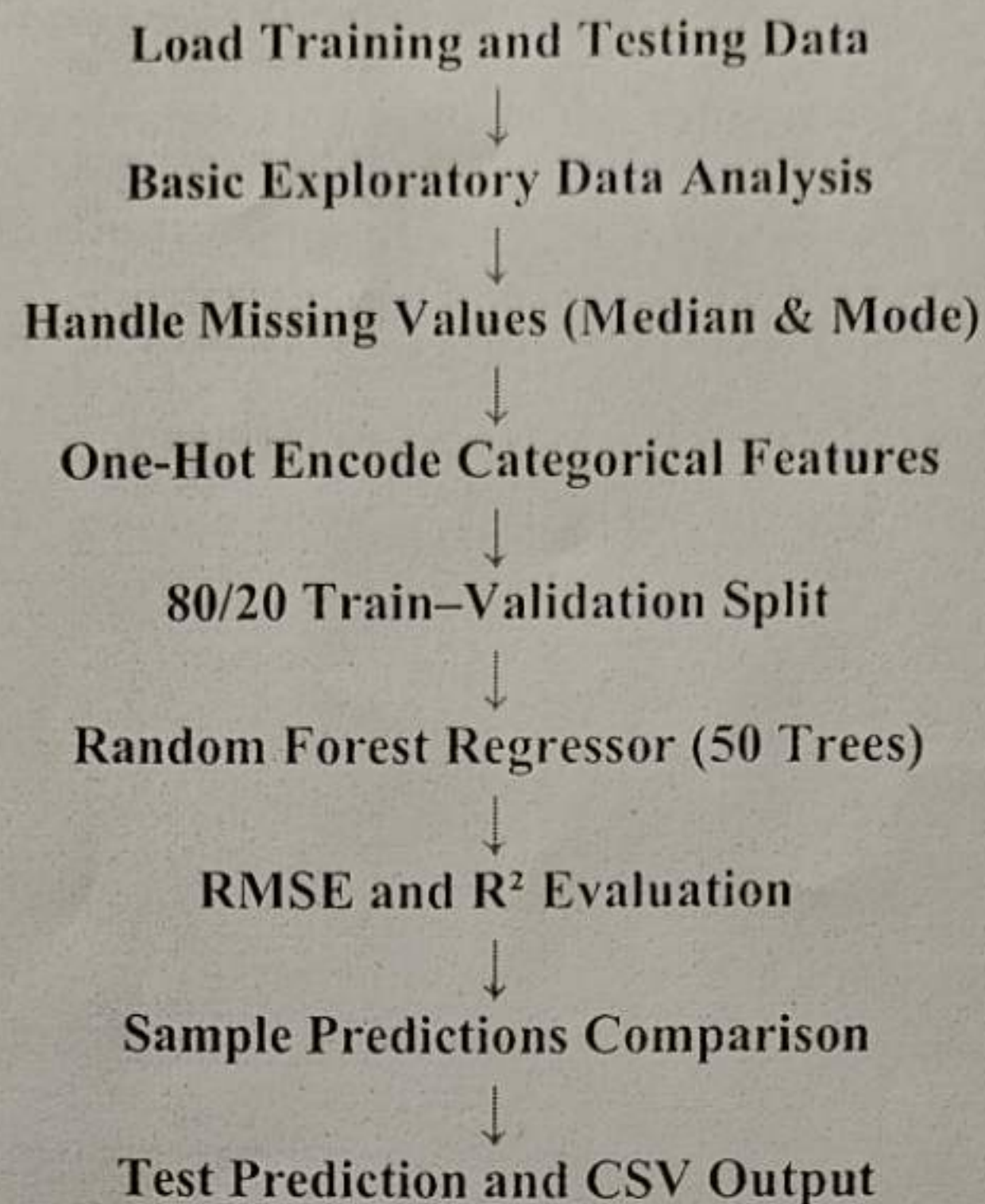
Missing categorical values are filled using the most frequent value (mode).

Categorical variables are then converted into numerical columns using one-hot encoding.

No manually constructed features are added to the dataset.

5. System Design

The proposed system follows a simple sequential machine learning pipeline.



5.1 Data Loading

The notebook automatically searches for train.csv and test.csv. The training data is separated into input features and the SalePrice target.

5.2 Exploratory Data Analysis

Basic EDA is performed before model training. The notebook displays the first five records, the ten columns with the largest number of missing values, descriptive statistics for SalePrice, and a histogram showing the distribution of house prices.

5.3 Basic Preprocessing

Numerical missing values are replaced with their column median. Categorical missing values are replaced with the column mode. After that, categorical variables are converted using one-hot encoding. No feature engineering is performed.

5.4 Model Training

The data is divided into training and validation sets using an 80/20 split with random_state=42.

The only model used is:

- Random Forest Regressor
- Number of trees: 50
- Random state: 42
- Parallel processing: enabled using all available CPU cores

5.5 Evaluation

Two evaluation metrics are calculated:

Root Mean Squared Error: $RMSE = \sqrt{[(1 / n) * \sum (y_i - \hat{y}_i)^2]}$

RMSE measures the typical prediction error in the same units as the house price.

Coefficient of Determination: $R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2]$

The notebook reports both RMSE and R^2 , with R^2 also displayed as a percentage for easier interpretation.

6. Results & Conclusion

6.1 Results

The final notebook produces the following outputs:

1. Dataset dimensions after loading.
2. Basic EDA including the first five rows, missing-value summary, descriptive statistics, and price distribution.
3. The processed feature matrix after missing-value handling and one-hot encoding.
4. RMSE on the validation dataset.
5. R^2 score and its percentage representation.
6. A sample table comparing actual and predicted house prices.
7. A submission.csv file containing predictions for the test dataset.

The exact RMSE and R^2 values depend on the dataset and execution environment and are generated directly when the notebook is run. They are intentionally not hard-coded into this report.

6.2 Sample Output Format

The notebook displays a sample prediction table in the following format:

Actual Price	Predicted Price
\$154,500	\$142,402
\$325,000	\$322,386
\$115,000	\$116,632
\$159,000	\$152,357
\$315,500	\$324,582
\$75,500	\$85,982
\$146,000	\$151,564
\$84,500	\$85,774
\$135,500	\$130,673
\$210,000	\$208,450

6.3 Conclusion

The microproject successfully demonstrates an end-to-end machine learning workflow for house price prediction.

The system begins with basic exploratory analysis, performs simple missing-value handling and categorical encoding, trains a single Random Forest regression model, evaluates its predictions using RMSE and R^2 , and generates predictions for unseen test data.

Compared with a complex ensemble-based implementation, the final system is considerably simpler and faster to execute. This makes it appropriate for demonstrating the fundamental stages of a machine learning project: data preparation, exploration, model training, evaluation, and prediction.

6.4 Future Scope

The system can be extended in future work by:

- Comparing multiple regression algorithms.
- Adding carefully selected feature engineering techniques.
- Using cross-validation for more robust evaluation.
- Performing hyperparameter tuning.
- Developing a web interface for interactive house price prediction.
- Deploying the trained model as a simple prediction service.

7. References

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